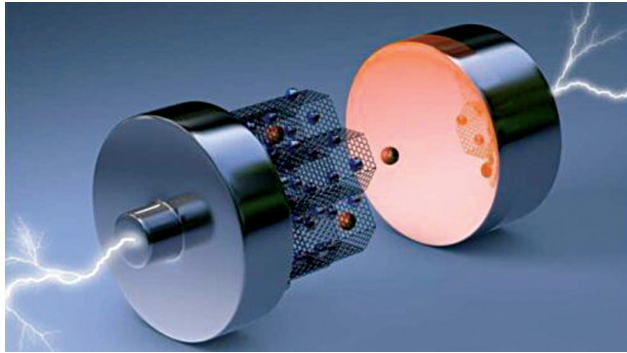


New Li-ion battery electrode enables faster charging

Researchers at the University of Texas at Austin have developed a new type of electrode for lithium-ion batteries that could provide greater power and faster charging. They used thin two-dimensional materials as building blocks of the



Magnetic field assists thick battery electrodes (Credit: The University of Texas at Austin)

electrode, arranged them to create thickness, and then used a magnetic field to manipulate their orientations. They used commercially available magnets during the fabrication process to stack the two-dimensional materials in a vertical alignment, creating a fast lane for ions to travel through the electrode. They were able to recharge the vertical thick electrode to 50% energy level in 30 minutes, compared with 2 hours and 30 minutes with the horizontal electrode.

Digital mask that ensures privacy during healthcare consultations

Scientists from Cambridge and China have developed a digital mask using three-dimensional (3D) reconstruc-

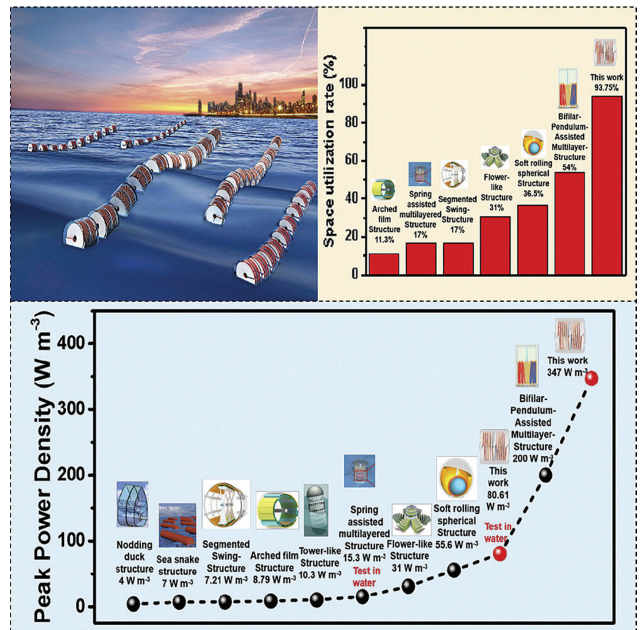


Digital mask for receiving features relevant to obtain medical data (Credit: <https://www.cam.ac.uk>)

tion and deep learning algorithms to remove identifiable features from facial images while keeping disease-relevant features needed for diagnosis. It takes an original video of a patient's face as input and provides a processed video as an output. This process removes the patient's personal biometric information as much as possible. It is very difficult to obtain original videos from digital mask videos as most of the essential information is removed during the process of conversion. This research will make telemedicine phone and video consultations possible, making healthcare delivery more efficient without compromising an individual's privacy concerns.

New device to generate static energy from waves

A team of researchers at the Chinese Academy of Sciences has invented a new device that could efficiently generate static electricity from waves. The device was configured with multiple small nanogenerators that can float on the surface of the ocean due to its lightweight. Each nanogen-



Anaconda-shaped spiral multi-layered triboelectric nanogenerators with ultra-high space efficiency for wave energy harvesting (Credit: <https://techxplore.com>)

erator is fixed together using a flexible connector. The nanogenerators consist of spring-like coils within them with parts that rub against one another as the device is moved by the waves. The touching parts generate static electricity, which is then pooled to be tapped and used as a power source. The device was producing 30 times better than other designs that were implemented to produce electricity from waves.